# Beyond the One-Hour Standard: Using Auxiliary Data to Reconstruct Nigeria's Discontinued Unemployment Series

by

72601

A Capstone Project submitted in partial fulfilment of the requirements for the degree of MSc Applied Social Data Science

at

Department of Methodology

London School of Economics and Political Science

Supervisor: Tom Robinson

July 2026

Estimated Word Count: 9954

**Abstract**

The adoption of the International Labour Organization (ILO) definition of unemployment and the consequent discontinuation of domestic measures in many developing countries have attracted substantial criticism, particularly around its limited representation of local labour-market conditions. While there have been theoretical debates on the utility of domestic definitions, few empirical studies have attempted to reconstruct them once they are officially discontinued.

This study addresses that gap by examining Nigeria's recent adoption of the ILO definition. Using machine-learning techniques, it seeks to reconstruct the discontinued domestic unemployment measure from available long-term data series. The research is motivated by the drastic change in reported unemployment rates following the adoption of the new definition.

To achieve this objective, the study compiles administrative and alternative data sources to estimate the previous measure of unemployment. It additionally compares the determinants of both definitions using model coefficients to assess whether their differences extend beyond the effect of arbitrary threshold changes. An elastic net model is adopted as the most appropriate modelling framework within the Nigeria context, specifically, for its unique feature selection and extrapolation abilities. The findings suggest that the previous unemployment definition can be reconstructed with reasonable accuracy for the years examined. However, validation for future periods is not possible because the measure has been discontinued.

More importantly, the results indicate that although both definitions share several common drivers, certain socioeconomic factors influence them differently. These differences suggest that changes in classification thresholds create structurally distinct unemployment metrics.

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# 1 Introduction

## 1.1 Background and Motivation

Since 1923, the International Conference of Labour Statisticians (ICLS), a technical body of the International Labour Organization (ILO), has worked to develop international guidelines that standardize how countries define concepts related to work, employment, and labour statistics [1]. While the conference has produced standards adopted widely across countries, the structural diversity of economies around the world often generates debates on their applicability (Brandolini et al. 2006; Guataquí and Taborda 2006; Baah-Boateng 2015; Alenda-Demoutiez and Mügge 2020). A clear illustration of these structural differences can be seen in Table 6, which shows the varying units of works countries use to define their minimum wages — some fixing an hourly rate, others a monthly rate, and some formally recognizing both. This variation is not merely administrative; since the ILO defines employment as work performed for pay or profit, differences in how pay is standardized across countries raise the recurring question of whether a single unemployment standard can be applied uniformly across such diverse labour economies.

Notably, there are persistent concerns regarding the applicability of international labour standards in developing economies characterized by high levels of informality, where the consequences of statistical mismeasurement are particularly severe (Byrne and Strobl 2004; Alenda-Demoutiez and Mügge 2020). Scholars and practitioners have extensively debated several core tenets of the standard employment definition to determine their validity in developing contexts. Key contentions include whether an "active job search" should be strictly required to classify an individual as unemployed, and whether the minimum working-hour thresholds accurately reflect labour market realities (Brandolini et al. 2006; Hussmanns 2007; Byrne and Strobl 2004). The validity of these debates have been acknowledge by the ILO and broader, composite measures of labour underutilization (LU1–LU4) have sprang up in recent years (International Labour Organization 2011; International Conference of Labour Statisticians 2023).

The differences in measures resulting from different definitions are significant to policy and national decision making. Nigeria's National Bureau of Statistics (NBS) provides a contemporary

[1] Table 7

case study where this can be explored, having maintained a domestically adapted unemployment definition for over two decade before fully transitioning to the ILO standard in 2023 (National Bureau of Statistics 2023; World Bank, n.d.). Prior to this methodological shift, Nigeria's NBS operated under a framework that initially classified individuals working less than 40 hours a week as unemployed (Kale and Doguwa 2015), a threshold that was later revised downward to 20 hours per week (National Bureau of Statistics 2021; Kale and Doguwa 2015). Under this stringent domestic framework, Nigeria recorded a headline unemployment rate of 33.3% in the fourth quarter of 2020 (World Bank, n.d.; National Bureau of Statistics 2021).

Following the transition to the ILO definition, which classifies individuals as employed if they worked for at least one hour for pay or profit during the reference week, the reported unemployment rate plummeted to 5.3% in the fourth quarter of 2022 (World Bank, n.d.; National Bureau of Statistics 2023). This drastic statistical decline was driven almost entirely by the definitional and methodological shift, rather than a substantive improvement in underlying labour market conditions during the intervening period.

Consequently, several stakeholders have publicly criticized the change of methodology and others have outrightly dismissed the new figures (Proshare Research 2023; Aro 2023; Daily Trust 2023). This is consistent with arguements in literature stating that domestically adapted methodologies are arguably better suited to the economic reality of developing labour markets, where informal subsistence work and chronic underemployment are pervasive (Baah-Boateng 2015; World Bank, n.d.).

Despite the widespread recognition that labour market fluctuations are intricately linked to structural macroeconomic indicators and non-traditional data source (Askitas and Zimmermann 2009; Choi 2009) that continues to be available, there has been no empirical attempt to systematically reconstruct the discontinued, domestically adapted employment series to preserve historical continuity.

## 1.2 Research Question and Scope

The 2023 adoption of the ILO's one-hour employment criterion marked a major break in Nigeria's labour statistics. While the transition improved alignment with international standards, it also resulted in the discontinuation of the unemployment series previously produced under Nigeria's domestic methodology. Consequently, researchers, policymakers, and the public can no longer directly compare contemporary unemployment figures with historical estimates constructed using the earlier framework.

This discontinuity creates two important challenges. First, it limits the ability to assess changes in labour market conditions over time because observed differences in unemployment rates may reflect methodological changes rather than genuine economic developments. Second, it raises broader questions regarding the suitability of international labour standards in economies characterised by high levels of informality and underemployment.

This study addresses these challenges by investigating whether auxiliary data sources can be used to reconstruct Nigeria's discontinued unemployment series. In particular, it examines whether administrative indicators and digital trace data can recover the variation previously captured by Nigeria's domestic unemployment definition.

The primary research question is:

*To what extent can auxiliary socioeconomic indicators and digital trace data accurately reconstruct Nigeria's discontinued unemployment series?*

To answer this question, the study considers three subsidiary questions:

Which socioeconomic and/or digital trace indicators are most predictive of unemployment under Nigeria's discontinued domestic definition?

What modelling framework can be used to learn the discontinued domestic unemployment definition?

How accurately can the machine learning model reconstruct historical unemployment rates using these indicators?

The study focuses on the period for which both unemployment statistics and auxiliary indicators are available, combining administrative macroeconomic data with digital search data. Rather than creating a new independent index for unemployment, the objective of this study is to assess whether discontinued unemployment rate series can be reconstructed in a statistically reliable manner and to explore what such reconstruction reveals about labour market distress in a high-informality economy.

By doing so, the study contributes to both the methodological literature on reconstructing discontinued statistical series and the broader debate regarding the measurement of unemployment in developing economies.

## 1.3 Contributions

This study contributes to the literature in four ways. First, it develops a framework for reconstructing labour market indicators following major methodological breaks in official statistics. While changes in statistical definitions are common, relatively little attention has been paid to how discontinued series can be systematically recovered to preserve historical continuity and enable longitudinal analysis. By combining macroeconomic indicators with recursive forecasting techniques, this study demonstrates a practical approach for extending discontinued labour market series beyond their official publication period.

Second, the study contributes new empirical evidence from a highly informal labour market context. Most unemployment nowcasting and reconstruction studies have been conducted in advanced economies characterized by formal labour markets and relatively stable statistical definitions. Nigeria provides a substantially different environment, where labour market participation is dominated by informal and survival-oriented activities, making it an important case for evaluating the robustness of reconstruction approaches.

Third, the study provides a direct comparison of the macroeconomic determinants associated with alternative unemployment definitions. Rather than treating unemployment as a fixed statistical construct, the analysis recognizes that different definitions capture different dimensions of labour market distress. Comparing the historical Nigerian definition with the ILO framework allows the study to examine whether different methodological definitions respond similarly to the broader economic conditions.

Finally, the study contributes to broader debates concerning statistical harmonization in developing economies. While internationally standardized indicators facilitate cross-country comparison, they may also obscure locally relevant dimensions of economic hardship. The findings therefore speak not only to unemployment measurement in Nigeria but also to wider questions concerning the balance between international comparability and domestic policy relevance in official statistics.

The remainder of this paper is structured as follows: Section 2 reviews the relevant literature, Section 3 describes the data and methodology, Section 4 presents the results, and Section 5 discusses the findings and concludes.

# 2 Literature Review

This section reviews the difficulties around operationalizing unemployment in developing countries alongside academic literature on the reconstruction and estimation of labour market indicators in contexts of sparse or incomplete data. It is structured in three main parts. First, it discusses debates on the adaptation of ILO standards in developing countries. Second, it reviews Nigeria as a contextual case study. Finally, it examines likely data sources and methodological frameworks to estimate unemployment rates.

## 2.1 Measuring Unemployment in Developing Economies

The conceptualization and measurement of unemployment have been subjects of intense historical and methodological debate. The earliest efforts to establish international statistical standards for unemployment trace back to 1895, when the International Statistical Institute in Berne attempted to address the non-comparability of unemployment statistics across industrialised European nations (Richardson et al. 1992). Through successive International Conferences of Labour Statisticians (ICLS) shown in Table 7, particularly the Eighth ICLS in 1954, the modern labour force framework was established. This framework officially defined the unemployed as individuals of working age who are simultaneously "without work," "currently available for work," and "actively seeking work" (Byrne and Strobl 2004; Richardson et al. 1992).

However, the strict application of these criteria generated persistent theoretical contention, leading scholars and national statisticians to question their validity in developing economies (Byrne and Strobl 2004). Historically, the statistical category of unemployment was constructed to measure labour distress in advanced, industrialized nations where formal wage-labour is the dominant social norm and where safety nets, such as unemployment insurance, allow individuals to visibly engage in active job searching (Benanav 2019; International Labour Organization 2011). In Low-Income Developing Countries (LIDCs), these structural conditions are fundamentally absent (Benanav 2019) as formal labour exchange channels are limited and comprehensive unemployment benefits are practically non-existent, the vast majority of the population cannot afford to remain completely idle (Baah-Boateng 2015; Morse 1970). Consequently, when formal jobs are scarce, workers are forced to take up low-productivity, informal survival activities or subsistence agriculture (Benanav 2019; Baah-Boateng 2015).

These structural realities have caused researchers and national statistical agencies to conclude that implementing domestically adapted definitions is not merely a deviation from international norms, but a statistically valid and necessary requirement for capturing true economic distress. As strictly applying the ILO's standard criteria in LIDCs can artificially deflate the unemployment rate, creating an "unemployment-informality trade-off" where high levels of precarious employment actively conceal true joblessness (Baah-Boateng 2015).

Several nations have historically adopted broad, domestic adaptations of unemployment definitions to correct these biases (Alenda-Demoutiez and Mügge 2020; Guataquí and Taborda 2006). For example, in early post-Apartheid South Africa, statisticians consciously defied international conventions to utilize an expanded unemployment indicator; this domestic adaptation was institutionally validated because it corrected the severe gendered and racialized biases of the strict ILO definition, thereby doing justice to the discouraged, marginalized Black population (Alenda-Demoutiez and Mügge 2020). Similarly, Trinidad and Tobago officially dropped the active job search requirement from its national definition, an adaptation empirically validated by research demonstrating that non-searching, marginally attached workers in developing nations possess the same transitional labour market behaviour as active job seekers (Guataquí and Taborda 2006; Byrne and Strobl 2004).

The theoretical validity of these domestic adaptations was ultimately recognized by the ILO itself. Acknowledging that the standard unemployment definition could be overly restrictive in certain labour market contexts, the 13th International Conference of Labour Statisticians (ICLS) in 1982 introduced a "relaxation provision" (Hussmanns 2007; Richardson et al. 1992). This provision explicitly permitted countries to relax the "seeking work" criterion in situations where labour markets were largely unorganized, where conventional job-search mechanisms were of limited relevance, or where self-employment constituted the dominant form of economic activity (International Labour Organization 1982; Richardson et al. 1992).

The ILO additionally addressed these limitations by argumenting the base unemployment measure and introducing several other labour untilization measures to help developing contexts characterized by widespread informality, precarious work, and underemployment (International Labour Organization 2011). These formal altercations and adjustments lend credibility to the need to having different unemployment definitions for less developing economies.

Despite this institutional flexibility and the acknowledgement that labour market realities differ substantially across countries, the political economy of global statistics — driven by international financial institutions, donor agencies, and the pursuit of cross-country comparability — has often encouraged developing nations to align with standardized international definitions. As a result, domestically adapted measures are sometimes abandoned in favour of harmonized indicators, even when local statisticians argue that such adaptations better reflect national labour market conditions (Alenda-Demoutiez and Mügge 2020).

## 2.2 Nigeria's Labour Market and Statistical Transition

Nigeria was historically one of the countries that actively developed a domestic adaptation of the unemployment definition and regularly reviewed it against the realities of its labour market (Kale and Doguwa 2015). In keeping with its statutory mandate, the National Bureau of

Statistics (NBS) periodically evaluated its statistical processes to ensure that they reflected the structural characteristics of the Nigerian economy while maintaining alignment with international best practices (Kale and Doguwa 2015).

This balancing act between international statistical standardization and local economic realities is not unique to Nigeria. Across the developing world, national statistical agencies have faced difficult choices regarding how labour market concepts should be adapted to account for informality, subsistence work, and weak formal labour market institutions. Different countries have resolved these tensions in different ways. India, for example, has long relied on a more complex "usual activity" framework to capture persistent underemployment and irregular labour force participation within its large informal sector. By contrast, South Africa moved in the opposite direction during the 1990s, replacing a broader unemployment measure with the stricter ILO standard in part to enhance international credibility, despite concerns that the change obscured labour market discouragement rooted in the legacy of apartheid (Alenda-Demoutiez and Mügge 2020).

Nigeria's experience sits within this broader debate between local relevance and international comparability. Like several other developing countries, Nigeria historically maintained a domestically adapted unemployment framework designed to better reflect the realities of its labour market. However, its eventual transition to the ILO standard represents one of the most significant methodological shifts observed in recent years.

Prior to 2014, the NBS classified individuals working fewer than 40 hours per week as unemployed (Kale and Doguwa 2015). Following a review of labour market conditions, the agency revised this framework to better reflect the growing importance of informal service and agricultural activities. Under the revised methodology, individuals working between 20 and 39 hours per week were classified as underemployed, while those working fewer than 20 hours were classified as unemployed (Kale and Doguwa 2015). This approach effectively used a relatively high work-hour threshold as a proxy for adequate labour market engagement in an economy where formal wage employment accounts for only a small share of total employment (World Bank, n.d.). Under this framework, Nigeria's headline unemployment rate reached 33.3% in the fourth quarter of 2020 (World Bank, n.d.; National Bureau of Statistics 2023).

Importantly, these domestic statistics did not exist in isolation from international comparisons. For many years, the International Labour Organization (ILO) produced modelled unemployment estimates for Nigeria alongside the country's official figures (International Labour Organization 2026). These estimates applied the ILO's standard labour force framework, which classifies individuals as employed if they perform at least one hour of work for pay or profit during the reference period. Designed to maximize international comparability and capture all forms of economic activity, including informal and part-time work, the ILO estimates consistently reported substantially lower unemployment rates for Nigeria, typically ranging between 3.7% and 5.7%.

The existence of these modelled estimates meant that internationally comparable unemployment indicators were already available prior to 2023. Nevertheless, in 2023 the NBS formally adopted the ILO methodology as its official unemployment measure and simultaneously introduced the broader labour underutilization framework, including LU2–LU4 indicators (National Bureau of Statistics 2023). While these measures provide additional information on labour market conditions, they are conceptually distinct from Nigeria's former domestic unemployment series.

As a result, the methodological transition did more than alter the official unemployment rate; it also discontinued the series that had previously been used to measure labour market distress under Nigeria's higher work-hour threshold. This created a break in the historical record, limiting the comparability of post-2023 unemployment figures with earlier estimates and creating a challenge for researchers seeking to evaluate long-run labour market trends.

## 2.3 Reconstructing the Unemployment Rate in Data-Sparse Contexts

The reconstruction of discontinued or missing macroeconomic series is a fundamental challenge in applied economics, particularly in developing nations where statistical agencies frequently alter methodologies or suffer from survey gaps. When a historical unemployment series is discontinued or measured infrequently, economists must estimate the missing data to preserve historical comparability and accurately evaluate long-term policy impacts (Vera-Jaramillo 2025). The academic literature on reconstructing the unemployment rate has evolved significantly, shifting from traditional econometric models anchored in macroeconomic theory to modern, computational frameworks that integrate high-frequency alternative data and machine learning. This section reviews the data sources, methodological arguments, and theoretical expectations that underpin the reconstruction of labor market indicators.

### 2.3.1 Macroeconomic Anchors and Structural Modeling

The traditional approach to estimating or forecasting missing unemployment data relies on established structural relationships within the macroeconomy. The foundational expectation is that labor market fluctuations do not occur in a vacuum; they are intrinsically linked to aggregate output and price levels. Researchers have historically utilized Okun's Law, which posits an inverse relationship between real GDP growth and unemployment, and the Phillips Curve, which describes the trade-off between inflation and unemployment, as primary theoretical anchors (Abiodun et al. 2025; Ball et al. 2019; Ojapinwa and Folorunso 2013). Because data such as the Consumer Price Index (CPI), Gross Domestic Product (GDP), and industrial production indices are routinely collected with higher frequency and consistency than labor force surveys, they serve as reliable covariates to model missing unemployment trends (Abiodun et al. 2025; Ball et al. 2019).

Furthermore, researchers argue that monetary policy transmission channels provide excellent leading indicators for labor absorption. In Nigeria, for example, composite leading indicators (CLI) utilizing high-frequency financial data — such as broad money supply (M2), nominal exchange rates, maximum lending rates, and inter-bank rates — have been proven to accurately trace and forecast unemployment cycles and turning points (Essien et al., n.d.; Tule et al. 2016). The expectation in structural modeling is that by capturing the broader macroeconomic environment, one can reliably estimate the trajectory of labor distress (Tule et al. 2016). However, the limitation of this approach is that administrative macroeconomic series often lack the granularity to capture sudden, high-frequency behavioral shifts in the labor market, particularly within the informal sector (Vera-Jaramillo 2025).

### 2.3.2 Digital Trace Data and Labour Market Measurement

The increasing availability of digital trace data has created new opportunities for measuring economic activity in near real time. Among the most widely studied sources of digital trace data are internet search queries, which record the information-seeking behaviour of individuals as they interact with search engines. A growing body of literature argues that changes in labour market conditions generate observable changes in online search activity, allowing search data to complement traditional labour market statistics (Askitas and Zimmermann 2009; Choi 2009).

The theoretical basis for this approach is behavioural. Individuals experiencing unemployment, underemployment, or economic insecurity often alter their information-seeking behaviour by searching for employment opportunities, government support programmes, income-generating activities, or other resources directly or indirectly associated with labour market distress. These behavioural responses create digital footprints that can be aggregated and analysed at scale (Pavlicek and Kristoufek 2015; Askitas and Zimmermann 2015).

Early studies demonstrated that internet search activity could improve forecasts of labour market indicators. Askitas and Zimmermann (2009) found that search queries related to employment significantly improved unemployment forecasting in Germany, while Choi and Varian (2009) showed that Google Trends data enhanced short-term prediction of unemployment claims in the United States. Subsequent research has extended these findings to a variety of contexts, suggesting that search activity can provide valuable information about labour market dynamics before official statistics become available (Pavlicek and Kristoufek 2015).

More recent studies have explored the application of digital trace data in developing economies. Evidence from countries such as Ghana and Turkey suggests that internet search activity may provide useful signals regarding labour market conditions, particularly when official statistics are infrequent or subject to publication delays (Adu et al. 2023; Gayaker and Türe 2025). These findings have contributed to a growing literature on economic nowcasting that combines traditional macroeconomic indicators with alternative digital data sources.

However, important limitations remain. Critics argue that internet search data is not derived from a probabilistic sample of the population and therefore suffers from significant selection bias (Mulero and Garcia-Hiernaux 2023). Search activity tends to overrepresent younger, urban, and more educated individuals while underrepresenting rural populations and groups with limited internet access. This concern is particularly acute in low-income developing economies where internet penetration remains uneven.

A second challenge relates to labour market structure. In economies characterized by extensive informal employment, traditional job-search mechanisms often play a limited role in labour market matching. Individuals engaged in subsistence agriculture, informal trading, or family enterprises may not rely on internet searches when seeking economic opportunities. Consequently, digital trace data may fail to capture important dimensions of labour market activity that are observable through household surveys (D'Amuri, n.d.; Mulero and Garcia-Hiernaux 2023).

The literature also highlights considerable variation in approaches to keyword selection. Researchers have employed strategies ranging from simple intuition-driven keywords to algorithmic expansion using related search terms, behavioural categorization, and machine learning-based feature construction (Smith 2016; Yi et al. 2021; Grybauskas et al. 2023). While no single approach has emerged as universally superior, the broader consensus suggests that careful feature selection is critical for extracting meaningful economic signals from search data.

Overall, existing evidence suggests that digital trace data can provide valuable supplementary information for labour market measurement. Nevertheless, relatively little research has examined diverse methodological approaches and whether such data can be used to reconstruct discontinued unemployment series in highly informal economies. The present study contributes to this literature by evaluating whether digital trace data, when combined with traditional socioeconomic indicators, can be used to reconstruct Nigeria's discontinued unemployment series and what keyword selection strategy can be further explored.

### 2.3.3 Modern Computational Frameworks: Temporal Disaggregation and Machine Learning

Recognizing the limitations of both low-frequency macroeconomic data and biased digital trace data, the frontier of labor market reconstruction has shifted toward integrated, multivariate machine learning frameworks. Modern researchers argue that accurate reconstruction requires combining diverse auxiliary datasets; ranging from macroeconomic indicators to passive spatial data, and enforcing strict statistical accounting identities (Vera-Jaramillo 2025).

To bridge the gap between annual aggregates and monthly fluctuations, scholars frequently employ temporal disaggregation techniques, such as the Chow-Lin or Denton methods, which use high-frequency auxiliary variables (like monthly exchange rates or CPI) to accurately interpolate missing intra-year labor data (Adu et al. 2023; Vera-Jaramillo 2025). Once the data is temporally aligned, supervised machine learning algorithms—such as Random Forests, XGBoost, and Ridge Regression—are deployed to capture the complex, non-linear dynamics between the auxiliary predictors and the labor market (Krishnamurthy, n.d.). For instance, a recent study reconstructing subnational labor indicators in Colombia successfully integrated temporal disaggregation with deep learning and tree-based models, utilizing a matrix of demographic, institutional, and macroeconomic covariates to estimate employment and informality in regions completely lacking direct survey coverage (Vera-Jaramillo 2025).

The prevailing expectation in contemporary literature is that ensemble machine learning models, when strictly calibrated against demographic benchmarks and macroeconomic realities, can effectively reconstruct discontinued or missing labor series (Krishnamurthy, n.d.; Vera-Jaramillo 2025).

# 3 Data and Methodology

This section outlines the empirical strategy and computational framework utilized to reconstruct the discontinued high-threshold unemployment series. It details the data selection process, the feature selection strategy, the choice of the predictive algorithm, and the evaluation framework.

## 3.1 Data Sources

### 3.1.1 NBS Labour Statistics

In the case Nigeria, there are two potential unemployment thresholds to reconstruct: the historical 40-hour benchmark or the 20-hour operational definition introduced during the 2014 methodological review (Kale and Doguwa 2015). From 1999 to 2014, the National Bureau of Statistics (NBS) exclusively reported the 40-hour definition. Following the 2014 revision, the NBS continued to dual-report the 40-hour data alongside the new 20-hour metric and total labour force figures, allowing for seamless conversion to the previous baseline (National Bureau of Statistics 2016; Kale and Doguwa 2015). Because the 40-hour threshold provides a consistent, unbroken metric across the entire historical span up to the 2020 discontinuation, it is selected as the preferred target variable to be modelled in this study.

### 3.1.2 Data Construction

The primary dataset consists of macroeconomic, monetary, and labour-market indicators collected from the National Bureau of Statistics (NBS) database and the Central Bank of Nigeria (CBN) official statistical database. Key variables include the Consumer Price Index (CPI), savings rate, prime and maximum lending rates, Gross Domestic Product (GDP), and their lagged values. Complementary UK indicators—such as GDP, inflation, and unemployment—were accessed via the Office for National Statistics (ONS) and the Bank of England, with some series retrieved through the Federal Reserve Bank of St. Louis's FRED database, which republishes ONS and Bank of England data for international comparability. These indicators were selected based on both theoretical and empirical evidence linking macroeconomic conditions, monetary policy, and labour market outcomes (Ojapinwa and Folorunso 2013; Tule et al. 2016; Umoru and Anyiwe 2013). They were further validated in modelling.

In addition to conventional socioeconomic indicators, the study explored the use of digital trace data obtained from Google Trends through the API. The inclusion of search query data was motivated by a growing literature demonstrating the value of digital behavioural signals for labour market nowcasting and unemployment forecasting (Choi 2009; Askitas and Zimmermann 2009;

Adu et al. 2023). Google trends data, particularly, was selected as it is the largest search engine and the only platform which provides access to its aggregated search data.

However, the Google Trends database only provides consistent observations from 2004 onward. Consequently, including these data in the modelling would require the exclusion of all earlier observations. Given that the primary objective of this study is the long-term reconstruction of a discontinued unemployment series, restricting the analysis to the post-2004 period would substantially reduce the available historical sample and limit the assessment of model performance across different economic cycles. Consequently, two modelling frameworks were considered: (1) a baseline framework relying exclusively on macroeconomic and labour-market indicators and (2) an extended framework incorporating digital trace variables. This approach allows the study to evaluate the incremental contribution of digital trace data while preserving the longer historical series required for reconstruction.

## 3.2 Data Processing

### 3.2.1 Automated Feature Selection and Dimensionality Reduction

The literature provides no universally accepted procedure for selecting unemployment-related search terms. Existing studies employ approaches ranging from manually selected keywords based on economic theory to large-scale algorithmic expansion using related search queries (Smith 2016; Yi et al. 2021). Given the exploratory nature of this study and the limited evidence available for Nigeria, a hybrid feature-construction framework was adopted.

The process consisted of two stages. In the first stage, a Large Language Model (LLM) was used to generate candidate search terms associated with unemployment, underemployment, job search behaviour, income insecurity, economic hardship, migration, and related dimensions of labour market distress. The objective was not to identify predictors directly but to construct a broad theoretical search space informed by existing literature and contextual knowledge of the Nigerian labour market.

In the second stage, Google Trends data were collected for the candidate terms and subjected to automated dimensionality reduction. Because search queries frequently exhibit high levels of collinearity and redundancy, feature selection and dimensionality reduction procedures were applied to retain only variables containing useful predictive information. Consistent with recommendations in the forecasting literature, regularization-based methods were employed to reduce model complexity and mitigate overfitting (Yi et al. 2021).

The resulting digital trace variables were integrated with traditional socioeconomic indicators, including macroeconomic and labour market measures, to create a unified modelling dataset. This combined dataset formed the basis for subsequent machine learning analyses.

### 3.2.2 Feature Engineering and Synchronizaation

In longitudinal macroeconomic analysis, variables such as the Consumer Price Index (CPI) and Gross Domestic Product (GDP) are periodically rebased by national statistical agencies to reflect updated consumption patterns and base-year prices. For this study, the constant-price GDP series (anchored to 2010 prices) was selected, utilizing the comprehensive backcasted series provided by the National Bureau of Statistics (NBS) to ensure historical continuity.

To mitigate the scaling artifacts inevitably introduced by such macroeconomic rebasing, all predictors are transformed into year-on-year (YoY) growth rates. Because percentage changes are mathematically scale-invariant, this YoY transformation structurally neutralizes the multiplicative level shifts inherent to base-year updates.

To ensure strict temporal synchronization between the predictors and the target variable, the historical unemployment rate is similarly transformed into YoY changes. Quarter-on-quarter (QoQ) differencing was specifically rejected for this reconstruction. Because portions of the historical unemployment series contain imputed intra-year observations, QoQ transformations risk capturing and amplifying the noise of these short-term imputations rather than actual economic shifts. Conversely, YoY differencing smooths out this imputed volatility, ensuring the model learns the true, underlying structural relationships of the labor market.

Specifically, this uniform YoY transformation circumvents the trending variable problem, which introduces severe risks of spurious regression (A et al. 2011; Rob J Hyndman and George Athanasopoulos, n.d.). Cumulative macroeconomic indicators, such as raw CPI and constant-price GDP, inherently exhibit strong, deterministic time trends. Training a predictive model on these raw, non-stationary levels forces the algorithm to learn shared temporal drift rather than true relationships, fundamentally destroying its out-of-sample reconstruction accuracy (Rob J Hyndman and George Athanasopoulos, n.d.). As established in the time series forecasting literature, applying first-order or seasonal differencing to achieve stationarity is a strict prerequisite for statistical validity (Krishnamurthy, n.d.; Rob J Hyndman and George Athanasopoulos, n.d.). Therefore, the training target is specified not as a raw level, but as the stationary, year-on-year change in the unemployment rate.

Let $Y_t$ denote the quarterly national unemployment rate at time $t$. The modeling target is defined as:

$$\Delta_4 Y_t = Y_t - Y_{t-4}$$

To maintain mathematical consistency across the system, the macroeconomic and financial predictor space is synchronized to match this exact integration order:

Real GDP Growth: Transformed into a year-on-year growth rate to strip out long-term economic drift and capture the true output gap:

$$\Delta_4 \text{GDP}_t = \frac{\text{GDP}_t - \text{GDP}_{t-4}}{\text{GDP}_{t-4}}$$

Monetary and Interest Rates: Financial rate metrics (e.g., Savings, Prime Lending, and Maximum Lending rates) are converted into year-on-year basis-point changes:

$$\Delta_4 R_t = R_t - R_{t-4}$$

Consumer Price Index (CPI): To prevent temporal aggregation bias, monthly CPI levels are aggregated to quarterly mean levels first, followed by a quarter-on-quarter (QoQ) growth calculation to capture high-frequency price momentum:

$$\text{CPI\_MoM}_t = \frac{\overline{\text{CPI}}_t - \overline{\text{CPI}}_t - 1}{\overline{\text{CPI}}_{t-1}}$$

### 3.2.3 Temporal Disaggregation and Data Leakage Considerations

The historical unemployment target series spanning 1999–2014 was originally compiled at an annual cadence and subsequently disaggregated into quarterly observations via mathematical interpolation. Consequently, within-year quarterly variations (e.g., $t-1$, $t-2$, $t-3$) during this period represent algorithmic artifacts rather than observed economic reality.To prevent the model from inadvertently learning and exploiting these artificial disaggregation patterns—which constitutes a severe form of data leakage—within-year autoregressive terms are strictly omitted from the predictor matrix. Autoregressive structure is restricted exclusively to cross-year historical boundaries representing true observed quarterly cycles: $Y_{t-4}$ (the 1-year lag) and $Y_{t-8}$ (the 2-year lag).

## 3.3 Modelling Framework

### 3.3.1 The Elastic Net

A critical methodological choice in this dissertation is the utilization of an Elastic Net regression model rather than tree-based machine learning algorithms or deeep learning frameworks. While ensemble methods like Random Forests are highly popular for their ability to map complex, non-linear relationships, they suffer from a fundamental mathematical limitation in time series forecasting: they cannot extrapolate beyond the range of target values observed in their training data. Because Nigeria's unemployment rate experienced unprecedented, continuous historical highs (reaching 33.3% before the methodology was discontinued), a Random Forest model would fail to accurately forecast rising unemployment rates that exceed historical maximums. Similarly, a deep learning model will require more data to learn a stable pattern compared to the quarterly data available in this case. To overcome these limitations, the Elastic Net model was selected. The Elastic Net is a regularized regression method that combines the penalties of both Ridge regression and the Least Absolute Shrinkage and Selection Operator (LASSO) (Olukanmi et al. 2021). This dual-penalty framework effectively handles the severe multicollinearity inherent in using multiple macroeconomic lags, performing continuous variable selection while shrinking less relevant coefficients to prevent overfitting. Crucially, because it retains a linear mathematical structure, the Elastic Net possesses the necessary capacity to accurately extrapolate and predict rising unemployment rates well beyond the limits of previously seen historical values.

## 3.4 Model Evaluation: Rolling Window Validation

To rigorously assess the predictive accuracy of the Elastic Net model, the study employs a rolling window evaluation approach (D'Amuri and Marcucci 2010). Standard randomized cross-validation is often mathematically invalid for time series data because it assumes observations are independent and identically distributed (i.i.d.), ignoring inherent temporal dependencies and evolutionary effects over time (Bergmeir and Benítez 2012).

Instead, the rolling window technique (also known as a rolling forecasting origin) preserves the chronological sequence of the data (Rob J Hyndman and George Athanasopoulos, n.d.). Under this approach, the model is trained on a fixed historical window of data (baseline window of $t_0$ quarters, 40% of all data in this case to give sufficient evaluation window) and is then used to predict the immediate out-of-sample periods ( one-quarter-ahead reconstruction performance ($t_0 + 1$)) (Bergmeir and Benítez 2012). The training window is then iteratively moved forward in time, and the model is re-estimated at each step (Rob J Hyndman and George Athanasopoulos, n.d.; Smith 2016). This procedure meticulously simulates the real-time constraints faced by economic forecasters, ensuring that no future information is ever leaked into the training set (Rob J Hyndman and George Athanasopoulos, n.d.). By generating a sequence of sequential out-of-sample predictions, the rolling window approach provides a highly robust, unbiased evaluation of the model's true accuracy across different economic cycles.

At each rolling origin step, an adaptive Elastic Net Regularization framework is deployed, minimizing the penalized residual sum of squares by balancing Ridge ($L_2$) and Lasso ($L_1$) penalties:

$$\min_{\beta_0,\beta} \left\{ \frac{1}{2N} \sum_{i=1}^{N} \left(y_i - \beta_0 - x_i^T \beta\right)^2 + \lambda \left[ \frac{1-\alpha}{2} \|\beta\|_2^2 + \alpha \|\beta\|_1 \right] \right\}$$

Rather than hardcoding the mixing parameter ($\alpha$), a dual-parameter hyperparameter optimization is executed at each rolling step. An internal cross-validation loop over five folds identifies the optimal shrinkage penalty ($\lambda_{\text{min}}$), while a simultaneous grid search across the alpha spectrum ($\alpha \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0\}$) evaluates and selects the precise blend of feature selection (Lasso) and collinearity smoothing (Ridge) that minimizes out-of-sample reconstruction error.

## 3.5 Level Reconstruction Mechanics

Because the Elastic Net model is trained on stationary, differenced data to preserve statistical integrity, a vital post-estimation step is required to generate the final historical level series. The model yields a predicted year-on-year change ($\Delta_4 \hat{Y}_t$). The actual reconstructed level of the National unemployment rate ($\hat{Y}_t$) is mathematically operationalized by integrating the predicted difference back onto the true historical anchor from one year prior ($Y_{t-4}$):

$$\hat{Y}_t = \Delta_4 \hat{Y}_t + Y_{t-4}$$

This recursive integration ensures that while the model learns from clean, trend-free innovations, the final output is delivered on the original percentage-point scale required for policy analysis and historical interpretation.

## 3.6 Predictor Importance from Elastic Net Coefficients

To identify the structural drivers of the reconstruction, the in-sample regression coefficients were examined. In addition, feature importance was estimated using Permutation Feature Importance across twenty random shuffles ($n_{sim} = 20$). Repeating the permutation procedure reduces random variation in the resulting importance estimates and improves their stability.

For each predictor, its values were randomly permuted, thereby breaking its relationship with the target variable while leaving all other features unchanged. The resulting increase in Root Mean Squared Error (RMSE) was used as a measure of the predictor's contribution to model performance.

# 4 Results

This chapter presents the results of the unemployment forecasting models developed using quarterly macroeconomic indicators and digital trace data for Nigeria. The analysis employed Elastic Net regression with rolling-origin cross-validation to predict unemployment rates one quarter ahead. Rolling-origin evaluation was chosen to preserve temporal ordering and avoid look-ahead bias commonly associated with conventional k-fold cross-validation in time series applications.

The feature set consisted of quarterly macroeconomic indicators, including inflation (CPI), GDP, savings rates, lending rates, and autoregressive unemployment terms. Alternative model specifications, including Google Trends data and a baseline model using lag-only features, were also estimated and reported to evaluate robustness.

## 4.1 Descriptive Analysis

The modelling dataset covered the period from January, 1999 to October, 2020, yielding 88 quarterly observations after converting the initial annual reported values into quarterly observations. The plot of the target variables are shown below;

### 4.1.1 Evolution of Unemployment under Different Definitions

Note: Pre-2014 values were temporally disaggregated from annual data using the Denton-Cholette procedure; from 2014 Q4 onward the values are computed from quarterly NLFS microdata. The shaded band marks the splice point.

Figure 1 shows a harmonized quarterly unemployment series for Nigeria between 1999 and 2020 constructed under a consistent less-than-40-hours-per-week definition of unemployment. To ensure comparability over time, observations collected under subsequent NBS methodologies (the 20-hour definition) were converted and temporally disaggregated to produce a unified series. The resulting estimates indicate a long-run increase in unemployment rate, with unemployment rising from below 10% in the late 1990s to over 50% by 2020. While the series exhibits periods of relative stability, particularly during the early and mid-2000s, unemploy-

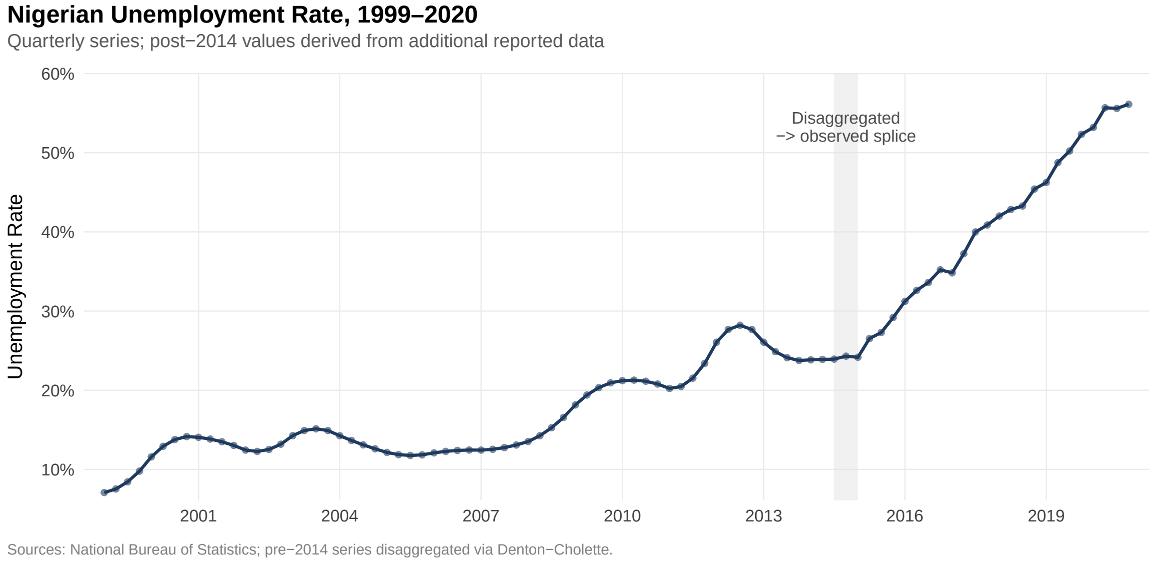


Figure 1: Quarterly unemployment rate under the 40-hour definition, 1999 Q1 to 2020 Q4.

ment increases substantially after 2015, reflecting the combined effects of economic slowdown, the 2016 recession, and broader structural labour market challenges. Unlike the ILO modelled unemployment series Figure 2, which remains below 6% throughout the same period, this harmonized measure captures the prevalence of insufficient labour market engagement under a substantially stricter employment threshold. The divergence between the two series illustrates how alternative definitions of employment can generate markedly different assessments of labour market conditions

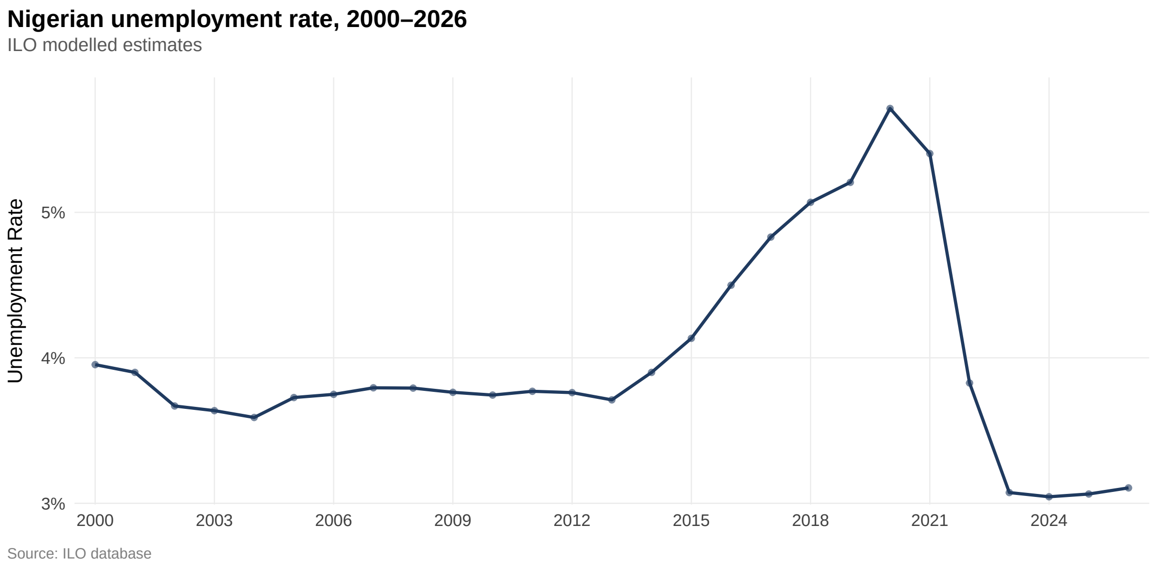


Figure 2: Yearly modelled unemployment rate under the ILO definition

Figure 2 presents the ILO modelled unemployment rate for Nigeria between 2000 and 2026. Unlike Nigeria's former domestic unemployment measure, these estimates are produced using the ILO's internationally harmonized labour force framework, which classifies individuals as employed if they perform at least one hour of work for pay or profit during the reference period. As a result, the series remains consistently low, ranging from approximately 3.1% to 5.7% over the period. The sharp contrast between these estimates and Nigeria's former domestic unemployment rates highlights the extent to which measured labour market conditions depend on the underlying definition of employment.

#### 4.1.2 Spurious Correlation Arising From Common Temporal Trends

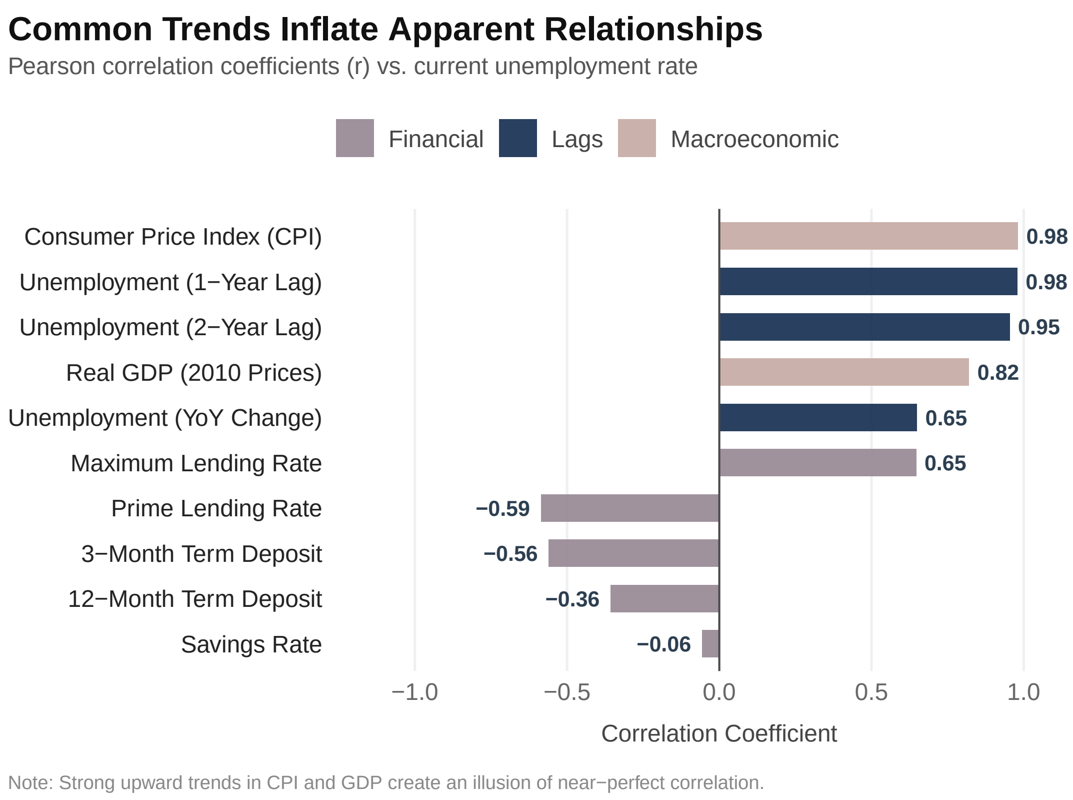


Figure 3: Raw Correlations of Features

Figure 3 reports pairwise correlations between unemployment and the candidate predictor variables using the original non-stationary series. Several variables exhibit remarkably strong correlations with unemployment. For example, CPI, GDP, and autoregressive features show correlations exceeding 0.8 in absolute value.

However, these relationships need to be carefully considered. Many macroeconomic variables exhibit strong time trends and persistence. When two unrelated variables trend upward over time, they may appear highly correlated even in the absence of a meaningful economic relationship. This phenomenon, commonly referred to as spurious correlation, creates temporal confounding in predictive modelling.

To distinguish genuine economic relationships from common temporal trends, I transformed the variables to achieve stationarity and the correlation analysis is repeated below. Following differencing, several previously strong relationships weakened substantially, while others remained relatively robust.

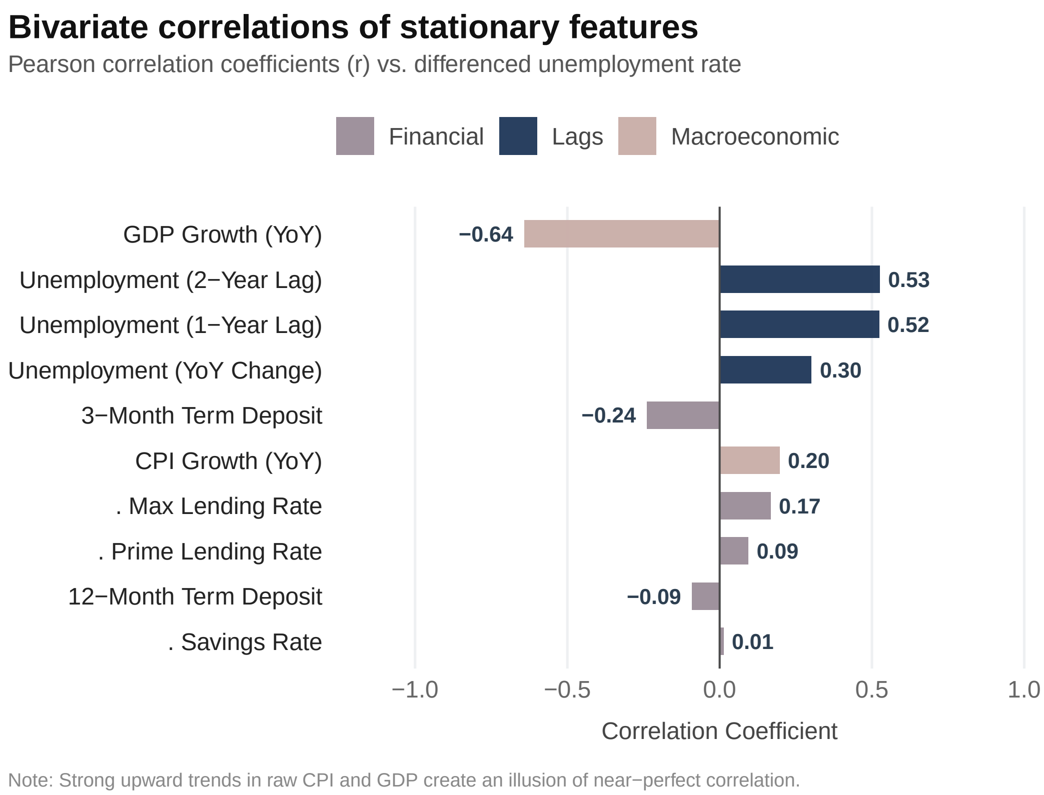


Figure 4: Stationary Correlations of Features

The comparison shows that much of the predictive power in the raw data reflects shared temporal dynamics rather than genuine economic relationships, so reconstruction models built on raw values risk producing spurious results. Nevertheless, a subset of variables continues to exhibit meaningful associations with unemployment after stationarity adjustments, suggesting that these indicators contain information beyond simple co-trending behaviour.

## 4.2 Model evaluation

### 4.2.1 Elastic Net Model Performance

The Elastic Net model was evaluated using rolling-origin cross-validation with an expanding training window.

Table 1 presents the forecasting performance metrics.

Table 1: Model Performance Summary

| Metric | value |
|---|---|
| RMSE | 2.765 |
| MAE | 2.280 |
| NRMSE | 0.080 |
| 90% Prediction Interval Coverage | 0.800 |

The model achieved an RMSE of 2.765 and an MAE of 2.28, indicating that forecasts were generally close to observed unemployment rates. The normalized RMSE of 0.08 indicates that the forecast errors amounted to approximately 8.05% of the average unemployment rate, suggesting relatively low error relative to the scale of the series.

The prediction intervals achieved a coverage rate of 80%, compared with the nominal 90% level. Although slightly below the target, this indicates that the uncertainty estimates were reasonably well calibrated.

### 4.2.2 Out-of-Sample Reconstruction Visualization

Figure 5 compares observed and predicted unemployment rates across the evaluation period.

The forecasts closely tracked medium-term movements in unemployment, although some deviations occurred during periods of rapid change. This pattern is expected because Elastic Net regression primarily captures systematic relationships rather than unexpected shocks.

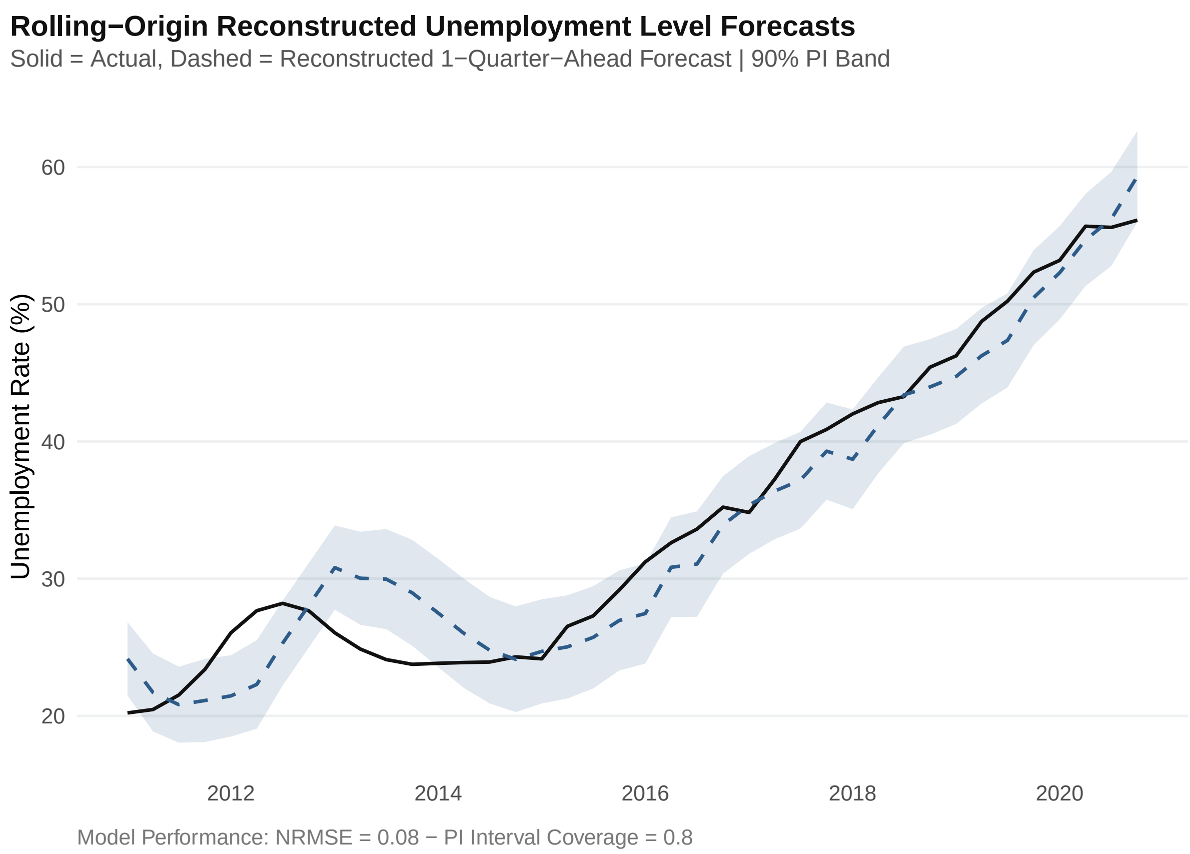


Figure 5: Rolling Out-of-Sample Reconstruction

#### 4.2.3 Digital Trace Extension

##### 4.2.3.1 Effect of Including Digital Trace Features

To assess whether digital trace data improved predictive performance, the model was re-estimated including Google Trends search volumes which where reduced to 2 principal components alongside the macroeconomic predictors.

Table 2: Extended Model Performance Summary

| Model | RMSE | MAE | NRMSE | 90% Prediction Interval Coverage |
|---|---|---|---|---|
| Full Elastic Net | 2.765 | 2.280 | 0.080 | 0.800 |
| Full Elastic Net + Trace | 2.276 | 1.773 | 0.059 | 0.821 |

The resulting model exhibited a modest improvement in predictive accuracy relative to the macroeconomic-only specification. However, this apparent improvement cannot be directly compared.

Unlike the benchmark model, the digital trace specification could only be estimated using data from 2004 onwards because Google Trends data only became available after 2004. Consequently, the inclusion of search data substantially reduced the length of the estimation and evaluation sample, producing a considerably shorter evaluation window which is arguably easier to estimate. Forecasting performance is therefore not directly comparable across the two specifications, as prediction difficulty generally depends on the length and historical coverage of the evaluation period.

The final forward projection reconstruction is based solely on the macroeconomic specification, allowing for a substantially longer estimation period while still delivering strong predictive accuracy.

#### 4.2.4 Comparison with Lag-Only Model

A naive forecast or reconstruction exercise will be based only on the immediate past values so that serves as a good bechmark to compare our model.

Table 3 compares forecasting accuracy.

Table 3: Model Comparison Summary

| Model | RMSE | MAE | NRMSE | 90% Prediction Interval Coverage |
|---|---|---|---|---|
| Lag Only | 3.366 | 2.759 | 1.024 | 0.739 |
| Full Elastic Net | 2.765 | 2.280 | 0.080 | 0.800 |
| Full Elastic Net + Trace | 2.276 | 1.773 | 0.059 | 0.821 |

As expected the model using only the one year and 2 year autoregressive features performs worse than the models with additional information.

## 4.3 Feature Importance Analysis

### 4.3.1 Determinants of the Domestic Definition

The coefficients of the model was used to identify the variables that contributed most to forecasting performance.

Figure 6 presents the relative importance of retained predictors of the domestic definition excluding the most dominant feature - GDP growth, to allow for better scaling of other features.

The bars are coloured according to coefficient sign, indicating whether increases in the predictor are associated with increases (positive) or decreases (negative) in predicted unemployment.

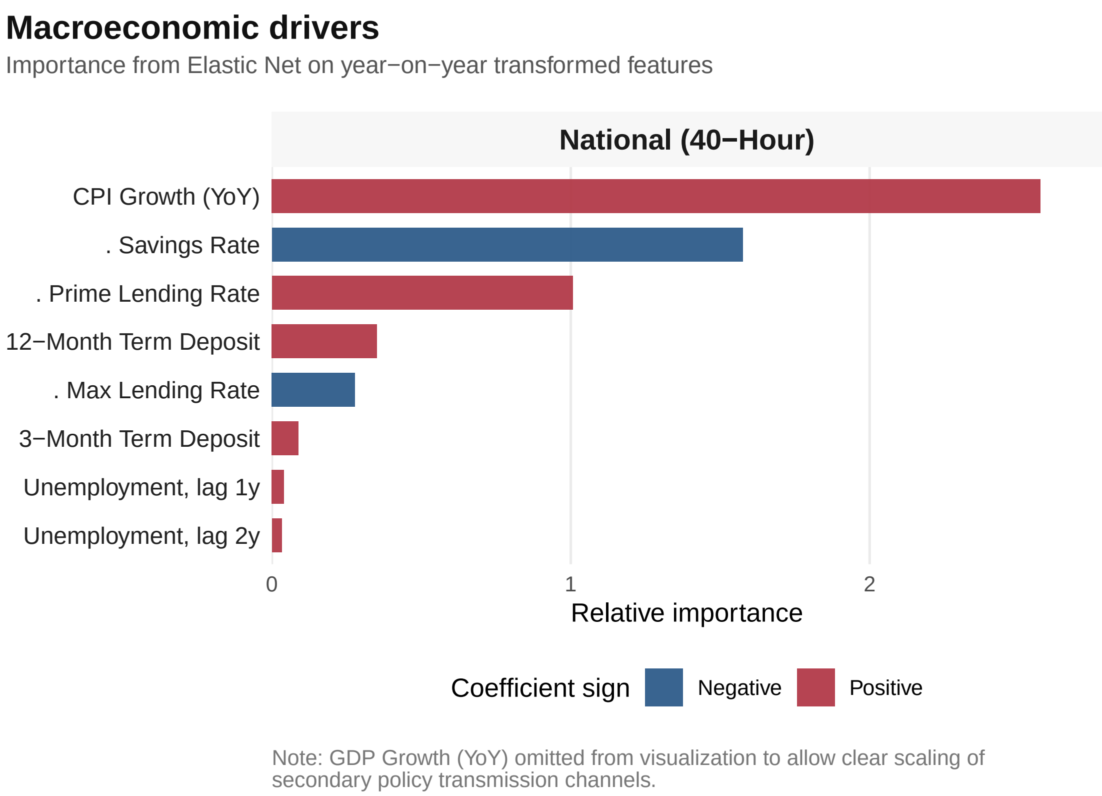


Figure 6: Feature Importances Using the Domestic Definition

Figure 6 presents the relative importance of the retained predictors for the domestic unemployment definition.

The domestic model suggests that, after GDP growth, inflation and savings behaviour are the most important macroeconomic predictors of unemployment. The domestic model places considerable importance on inflation alongside savings rates and lending conditions, suggesting that the historical unemployment definition was closely associated with broader macroeconomic conditions beyond changes in output alone. This is consistent with the conceptual basis of the 40-hour definition, which classified individuals working fewer than 40 hours per week as unemployed. Because this measure was intended to capture inadequate labour utilisation rather than complete exclusion from work, it is plausible that it responded more strongly to macroeconomic conditions that affect the availability and intensity of work. Financial variables, including savings and lending rates, therefore provide additional predictive information beyond GDP growth. In contrast, the relatively low importance of the autoregressive terms indicates that, once these macroeconomic conditions are accounted for, historical unemployment contributes comparatively little additional information to forecasting performance.

#### 4.3.2 Determinants of the ILO Definition

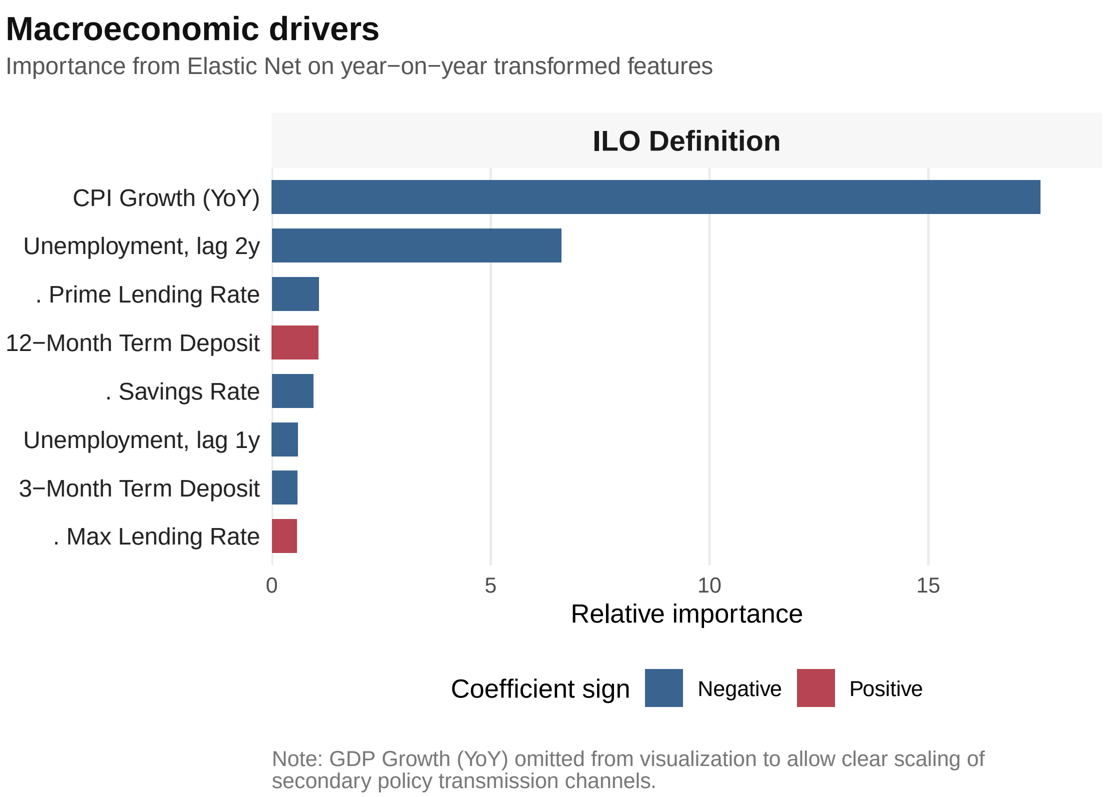


Figure 7: Feature Importance Using ILO Definition

Figure 7 presents the relative importance of the retained predictors for the ILO unemployment definition. As with the domestic model, GDP growth is omitted from the visualization because it dominated the importance rankings, allowing clearer comparison of the remaining predictors.

Unlike the domestic definition, the ILO model is overwhelmingly dominated by inflation, with the remaining variables contributing substantially less predictive information. Among the retained predictors, the two-year unemployment lag is the second most important variable, indicating greater persistence in the ILO unemployment series. Monetary variables, including lending rates and savings rates, remain informative but play a comparatively smaller role. This pattern is consistent with the conceptual basis of the ILO framework, which defines employment using a one-hour threshold and therefore primarily captures transitions into and out of employment rather than changes in work intensity. Consequently, the ILO measure appears to evolve more gradually over time and is less sensitive to broader macroeconomic fluctuations than the historical domestic definition.

### 4.3.3 Comparative Feature Importance

Table 4 compares the macroeconomic predictors retained by the Elastic Net models under the historical Nigerian and ILO unemployment definitions. Because Elastic Net performs variable selection through coefficient shrinkage, only variables with non-zero coefficients at the optimal penalty parameter are retained. Variables whose coefficients were shrunk to zero were excluded, indicating that they provided little additional predictive information after accounting for the remaining predictors.

Table 4: Comparative Feature Importance of Definitions

| Variable | Domestic Rank | ILO Rank |
|---|---|---|
| GDP Growth Rate (YoY) | 1 (Negative) | 1 (Negative) |
| Inflation Rate (CPI YoY) | 2 (Positive) | 2 (Negative) |
| Δ Savings Rate | 3 (Negative) | 6 (Negative) |
| Δ Prime Lending Rate | 4 (Positive) | 4 (Negative) |
| 12-Month Term Deposit Rate (Level) | 5 (Positive) | 5 (Positive) |
| Δ Maximum Lending Rate | 6 (Negative) | 9 (Positive) |

Table 4: Comparative Feature Importance of Definitions

| Variable | Domestic Rank | ILO Rank |
| --- | --- | --- |
| 3-Month Term Deposit Rate (Level) | 7 (Positive) | 8 (Negative) |
| Autoregressive Lag (t-4, 1-Year) | 8 (Positive) | 7 (Negative) |
| Autoregressive Lag (t-8, 2-Year) | 9 (Positive) | 3 (Negative) |

GDP growth is the dominant predictor under both unemployment definitions, but the remaining feature rankings show clear structural differences. Under the domestic definition, inflation and monetary variables—including savings and lending rates—play a stronger predictive role. The ILO model relies more heavily on lagged unemployment, especially the two-year term, indicating greater persistence.

These contrasts reflect the definitions themselves. Nigeria's historical measure classifies individuals working fewer than 40 hours per week as unemployed, making it sensitive to reductions in work intensity. The ILO standard counts anyone performing at least one hour of economic activity as employed, so declines in hours or earnings that do not eliminate work entirely affect the domestic measure more strongly. In Nigeria's informal labour market, where workers rarely exit economic activity, adverse conditions typically push individuals into lower-quality or lower-hour jobs rather than full unemployment. The stronger role of inflation and financial variables under the domestic definition therefore aligns with a measure designed to capture inadequate labour utilisation rather than outright joblessness.

Coefficient signs reinforce these differences. Inflation increases unemployment under the ILO definition but reduces it under the domestic measure. In Nigeria, inflation—often driven by supply-side pressures—tends to push households into informal or survival-oriented work, raising the likelihood of meeting the ILO's one-hour threshold while still falling short of adequate employment. Lending rates show a similar reversal: tighter credit raises unemployment under the domestic definition but reduces it under the ILO measure, likely reflecting movement into informal self-employment when formal credit access deteriorates. These sign differences demonstrate that statistical definitions shape observed macroeconomic relationships.

These findings should be interpreted as predictive rather than causal. Feature importance reflects variables that improve out-of-sample forecasting, and coefficient signs indicate fitted relationships. Overall, the Domestic (40-hour) and ILO (1-hour) series respond differently to identical macroeconomic conditions and cannot be treated as interchangeable indicators of labour market distress.

#### 4.3.4 Feature Importance in a developed economy

To further explore whether the macroeconomic determinants of the ILO unemployment definition differ between developed and developing economies, the a similar set of macroeconomic predictors was used to model the United Kingdom's harmonised unemployment series. The United Kingdom was selected as a developed-economy benchmark to provide a direct comparison with the Nigerian ILO model.

Table 5: UK Harmonised Unemployment: Elastic Net feature importance and coefficient direction.

| Rank | Variable | Coefficient | Rel. Importance | Sign |
|---|---|---|---|---|
| 1 | Inflation (CPI YoY) | 15.3902 | 100.0 | Positive |
| 2 | GDP Growth (YoY) | -6.0264 | 39.2 | Negative |
| 3 | Unemployment Momentum (YoY) | 0.2142 | 1.4 | Positive |
| 4 | Δ Short Rate | -0.1333 | 0.9 | Negative |
| 5 | AR Lag (2-year) | -0.1027 | 0.7 | Negative |
| 6 | Δ Policy Rate | -0.0832 | 0.5 | Negative |
| 7 | Δ Long Rate | -0.0787 | 0.5 | Negative |
| 8 | AR Lag (1-year) | -0.0705 | 0.5 | Negative |

Table 5 presents the variables retained by the Elastic Net model for the United Kingdom's harmonised unemployment series. Inflation (CPI year-on-year growth) emerged as the dominant predictor, contributing substantially more to forecasting performance than any other variable. GDP growth ranked second with a negative coefficient, consistent with the expectation that stronger economic growth is associated with lower unemployment. The remaining predictors, including unemployment momentum, interest rate changes, and autoregressive lags, contributed relatively little additional predictive information after accounting for inflation and output growth. The predominance of inflation is consistent with labour markets in advanced economies, where price pressures are more closely linked to labour market tightness and aggregate demand. Unlike the Nigerian models, persistence plays a comparatively minor role, suggesting that current macroeconomic conditions explain a larger share of short-term movements in the UK's harmonised unemployment rate.

## 4.4 Reconstruction of the Discontinued Series

The fitted model is now used to project forward the discontinued series;

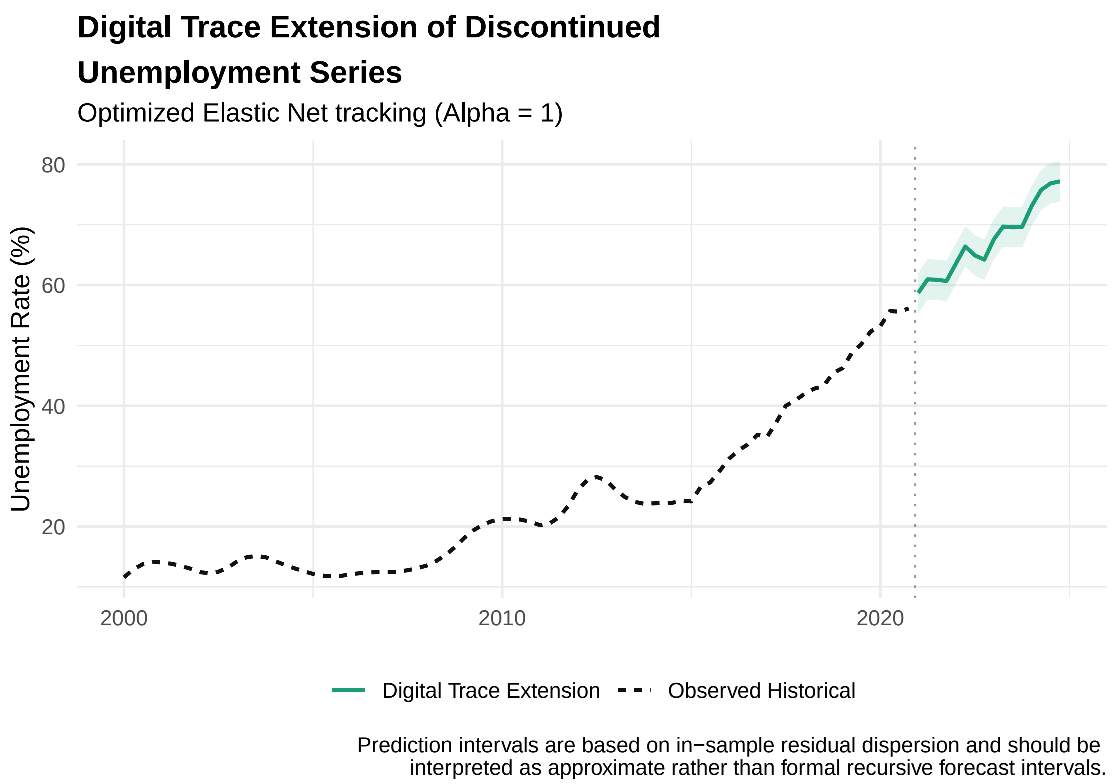


Figure 8: Forward Projection of Discontinued Series

The reconstructed series can be interpreted as a counterfactual continuation of Nigeria's historical unemployment framework. Rather than estimating a hypothetical labour market outcome, the reconstruction seeks to answer a specific measurement question: what would reported unemployment have looked like had the historical 40-hour definition remained in use after 2020?

The reconstructed series suggests that considerably higher levels of labour market distress would have been recorded under the previous framework, as expected. Due to limitations in the reconstruction—specifically the use of the initial 40-hour definition rather than the more current 20-hour one—the series is projected to rise as high as 77% when forward-projected, which may appear out of place initially. However, considering that the Nigerian Bureau of Statistics reported informal employment among Nigerians at 92.6% in the first quarter of 2023, and that this rate has consistently remained significantly high, the projection becomes plausible.

Importantly, this does not imply that either measure is inherently incorrect. Rather, each measure reflects a different conceptualization of meaningful economic participation. The historical Nigerian framework emphasized adequate labour engagement through a higher work-hour threshold, whereas the ILO framework prioritizes international comparability through a broader definition of employment. The reconstruction therefore illustrates the practical consequences of adopting one measurement philosophy over another.

# 5 Discussion

## 5.1 Can the Discontinued Series Be Reliably Reconstructed?

The results demonstrate that the discontinued unemployment series can be reconstructed with a reasonable degree of accuracy using macroeconomic indicators and autoregressive dynamics. Across the evaluation period, the Elastic Net model consistently outperformed the lag-only benchmark, indicating that macroeconomic conditions contain information beyond the persistence typically observed in unemployment series. This suggests that reconstruction is feasible and that discontinued official statistics need not necessarily create a permanent break in longitudinal labour market analysis.

However, the findings also illustrate the limitations of historical reconstruction. The recursive forecasting procedure relies on its own previous predictions, meaning that any prediction error propagates through subsequent forecasts. Although the reconstructed trajectory remains economically plausible over the evaluation period, forecast uncertainty inevitably increases as the prediction horizon extends. In practice, this implies that reconstruction is most reliable over relatively short-to-medium horizons unless the model is periodically recalibrated using newly observed unemployment data.

This limitation is particularly important in the context of the Nigerian unemployment series. Because the historical 40-hour definition has been permanently discontinued, there are no future observations against which reconstructed values can be validated. Unlike conventional forecasting problems, there is no opportunity to update the model or assess whether structural changes in the economy have altered the underlying relationship between macroeconomic variables and unemployment. Consequently, the reconstructed series should be interpreted as a statistically informed counterfactual rather than a replacement for future official statistics. Its primary value lies in preserving historical continuity and supporting longitudinal analysis, rather than providing definitive estimates of future unemployment under the discontinued definition.

## 5.2 What Do Digital Trace Data Contribute?

The inclusion of Google Trends variables produced a modest improvement in predictive performance, suggesting that behavioural search activity contains additional information relevant to labour market conditions. This finding is broadly consistent with the digital nowcasting literature, which argues that individuals experiencing labour market distress often alter their online behaviour before these changes become visible in official statistics.

However, the practical value of these gains must be interpreted in light of an important methodological trade-off. Because Google Trends data are only available from 2004 onwards, incorporating digital trace variables substantially shortened both the estimation period and the out-of-sample evaluation window. As a result, the predictive performance of the digital trace models is not directly comparable with the benchmark specification, since forecasting over a shorter and more recent period is generally less challenging than reconstructing a longer historical series spanning multiple economic regimes.

Where the objective is the historical reconstruction of discontinued official statistics, maximizing temporal coverage may be more valuable than obtaining marginal improvements in predictive accuracy. The exclusion of digital trace variables from the final reconstruction model was a methodological choice to use a model with a more extensive evaluation window.

## 5.3 What Do Different Unemployment Definitions Measure?

One of the most important findings of this study is that the historical Nigerian unemployment measure and the ILO unemployment measure capture different dimensions of labour market performance. As shown in Table 4, although GDP growth emerged as the dominant predictor under both definitions, the remaining feature rankings differ substantially. The historical domestic definition places greater emphasis on inflation and monetary variables, whereas the ILO definition relies more heavily on lagged unemployment, particularly the two-year autoregressive term, indicating greater temporal persistence. These differences suggest that changing the statistical definition alters not only the reported unemployment rate but also the macroeconomic relationships underlying the measure.

This divergence is consistent with the conceptual basis of the two definitions. The historical Nigerian framework classified individuals working fewer than 40 hours per week as unemployed, making it sensitive to changes in work intensity and employment adequacy. By contrast, the ILO framework classifies anyone performing at least one hour of work for pay or profit as employed, irrespective of whether that work provides sufficient income or stable employment. In an economy dominated by informal and necessity-driven work, adverse macroeconomic conditions are therefore more likely to reduce hours worked or earnings than to remove individuals from economic activity altogether. Consequently, the historical measure appears to capture broader labour underutilisation, while the ILO definition primarily reflects complete labour market exclusion.

To examine whether these relationships are specific to Nigeria (or other developing economies) or characteristic of the ILO definition more generally, the same modelling framework was applied to the United Kingdom's harmonised unemployment series. Table 5 shows that, while

GDP growth remained negatively associated with unemployment, inflation overwhelmingly dominated predictive performance, contributing more than twice the importance of GDP growth. By comparison, inflation played a comparatively modest role under both Nigerian definitions (Table 4). Furthermore, the Nigerian ILO model exhibited substantially stronger dependence on lagged unemployment than its UK counterpart, while several monetary variables changed both their relative importance and coefficient direction across countries.

These findings suggest that adopting a common statistical definition does not necessarily produce economically comparable unemployment measures. Although Nigeria and the United Kingdom both report unemployment using the internationally harmonised ILO framework, Table 4 and Table 5 demonstrate that the macroeconomic relationships underlying the measure remain strongly influenced by the structure of each country's labour market. In advanced economies, where unemployment primarily reflects movements into and out of formal wage employment, inflation and business cycle conditions play a more prominent role. In contrast, Nigeria's large informal sector means that economic shocks are more likely to affect employment quality and work intensity than complete labour market participation. Consequently, statistical harmonisation improves definitional consistency but does not necessarily harmonise the underlying economic phenomenon being measured.

Taken together, these results reinforce the argument that unemployment statistics are not purely objective measures but operational constructs shaped by statistical definitions and labour market institutions. Nigeria's transition to the ILO framework therefore represents more than a methodological change; it changes the aspect of labour market distress being quantified and, by extension, the macroeconomic relationships that researchers and policymakers observe.

## 5.4 Limitations

This study should be interpreted in light of several limitations relating to both the underlying data and the reconstruction methodology.

First, the reconstructed series represents a continuation of a historical statistical definition rather than a direct measure of contemporary unemployment. Although the study reconstructs a consistent 40-hour unemployment series, the National Bureau of Statistics (NBS) revised its domestic methodology in 2014 after determining that the previous threshold no longer adequately reflected the realities of Nigeria's evolving labour market. The revised framework lowered the unemployment threshold to 20 hours per week and introduced a distinct underemployment category for individuals working between 20 and 39 hours. Consequently, the post-2014 40-hour series used in this study is not directly observed but derived from the published unemployment and underemployment statistics to preserve consistency with the historical definition. While this harmonization enables longitudinal reconstruction, it implicitly assumes that the historical 40-hour threshold remained an appropriate benchmark after 2014, an assumption that may not fully reflect subsequent structural changes in the Nigerian economy.

Second, the modelling framework assumes that the historical relationship between unemployment and the selected macroeconomic indicators remains sufficiently stable over time. Although the model demonstrated good predictive performance during the validation period, this assumption may weaken as the economy evolves. Structural changes in labour market institutions, monetary policy, or employment composition could alter these relationships, leading to forecast drift. Under normal circumstances, such models would be periodically recalibrated using newly observed unemployment data. However, because the historical unemployment definition has been permanently discontinued, no future observations exist against which the reconstructed series can be validated or the model updated.

Finally, the study is constrained by the availability of explanatory variables and the relatively low frequency of the historical data. Because the reconstructed unemployment series is available only at a quarterly frequency, the analysis cannot capture higher-frequency labour market dynamics or short-term fluctuations that may occur within quarters. The analysis also relies primarily on aggregate macroeconomic indicators, which cannot fully capture regional, sectoral, or demographic heterogeneity in labour market conditions. Although behavioural digital trace data produced modest improvements in predictive performance, their inclusion substantially shortened the estimation period because Google Trends data are only available from 2004 onwards. Given the study's objective of reconstructing the longest possible historical series, preserving temporal coverage was prioritised over marginal improvements in predictive accuracy.

These limitations suggest that the reconstructed series should be interpreted as a statistically informed counterfactual that preserves historical continuity rather than a definitive estimate of the unemployment rate that would have been officially reported. Nevertheless, the framework provides a transparent and empirically validated approach for recovering discontinued statistical series, particularly in settings where methodological changes interrupt long-term economic analysis.

# 6 References

A, Sodipe, Ogunrinola, and Isaiah Olurinola. 2011. "Employment and Economic Growth Nexus in Nigeria." *International Journal of Business and Social Science* 2 (January): 232–39.

Abiodun, Adekoya, Magaji Sule, and Yahaya Ismail. 2025. *Empirical Analysis Of The Impact Of Unemployment On Economic Growth In Nigeria*. 13 (May): 63–80. https://doi.org/10.5281/zenodo.15311427.

Adu, Williams Kwasi, Peter Appiahene, and Stephen Afrifa. 2023. "VAR, ARIMAX and ARIMA Models for Nowcasting Unemployment Rate in Ghana Using Google Trends." *Journal of Electrical Systems and Information Technology* 10 (1): 12. https://doi.org/10.1186/s43067-023-00078-1.

Alenda-Demoutiez, Juliette, and Daniel Mügge. 2020. "The Lure of Ill-Fitting Unemployment Statistics: How South Africa's Discouraged Work Seekers Disappeared From the Unemployment Rate." *New Political Economy* 25 (4): 590–606. https://doi.org/10.1080/13563467.2019.1613355.

Aro, Busola. 2023. "'I Resisted It for 10 Years' — Yemi Kale Faults New Methodology for Unemployment Data." *TheCable*, August.

Askitas, Nikolaos, and Klaus F Zimmermann. 2015. *The Internet as a Data Source for Advancement in Social Sciences*.

Askitas, Nikos, and Klaus F. Zimmermann. 2009. "Google Econometrics and Unemployment Forecasting." *SSRN Electronic Journal*, ahead of print, June. https://doi.org/10.2139/SSRN.1480251.

Baah-Boateng, William. 2015. "Unemployment in Africa." *International Journal of Manpower* (Bradford, United Kingdom) 36 (5): 650–67. https://doi.org/10.1108/IJM-02-2014-0047.

Ball, Laurence, Davide Furceri, Daniel Leigh, and Prakash Loungani. 2019. "Does One Law Fit All? Cross-Country Evidence on Okun's Law." *Open Economies Review* 30 (5): 841–74. https://doi.org/10.1007/s11079-019-09549-3.

Benanav, Aaron. 2019. "The Origins of Informality: The ILO at the Limit of the Concept of Unemployment." *Journal of Global History* 14 (1): 107–25. https://doi.org/10.1017/S1740022818000372.

Bergmeir, Christoph, and José M. Benítez. 2012. "On the Use of Cross-Validation for Time Series Predictor Evaluation." *Information Sciences*, Data Mining for Software Trustworthiness, vol. 191 (May): 192–213. https://doi.org/10.1016/j.ins.2011.12.028.

Brandolini, Andrea, Piero Cipollone, and Eliana Viviano. 2006. "Does the Ilo Definition Capture All Unemployment?" *Journal of the European Economic Association* 4 (1): 153–79. https://doi.org/10.1162/jeea.2006.4.1.153.

Byrne, David, and Eric Strobl. 2004. "Defining Unemployment in Developing Countries: Evidence from Trinidad and Tobago." *Journal of Development Economics* 73 (1): 465–76. https://doi.org/10.1016/j.jdeveco.2002.12.005.

Choi, Hyunyoung. 2009. "Predicting Initial Claims for Unemployment Benefits." July.

D'Amuri, Francesco. n.d. *Predicting Unemployment in Short Samples with Internet Job Search Query Data*.

D'Amuri, Francesco, and Juri Marcucci. 2010. "'Google It!' Forecasting the US Unemployment Rate with A Google Job Search Index." *SSRN Electronic Journal*, ahead of print. https://doi.org/10.2139/ssrn.1594132.

Daily Trust. 2023. "'It Doesn't Make Sense', Ex-Statistician-General Faults NBS..." *Daily Trust*, August.

Essien, Sunday N, Garba A Manya, Mary O A Arigo, et al. n.d. *Monetary Policy and Unemployment in Nigeria: Is There a Dynamic Relationship?*

Gayaker, Savaş, and Hasan Türe. 2025. "National Unemployment Rate Forecast with Google Trends." *Mehmet Akif Ersoy Üniversitesi İktisadi Ve İdari Bilimler Fakültesi Dergisi* 12 (1): 107–23. https://doi.org/10.30798/makuiibf.1446639.

Grybauskas, Andrius, Vaida Pilinkienė, Mantas Lukauskas, Alina Stundžienė, and Jurgita Bruneckienė. 2023. "Nowcasting Unemployment Using Neural Networks and Multi-Dimensional Google Trends Data." *Economies* 11 (5). https://doi.org/10.3390/economies11050130.

Guataquí, Juan C., and Rodrigo Taborda. 2006. "Theoretical and Empirical Implications of the New Definition of Unemployment in Colombia." *Revista de Economía Del Rosario* 9 (1): 21–38.

Hussmanns, Ralf. 2007. "Measurement of Employment, Unemployment and Underemployment – Current International Standards and Issues in Their Application." *Bulletin of Labour Statistics* (Geneva, SWITZERLAND), no. 1 (January): vii–xxxi.

International Conference of Labour Statisticians. 2023. *Resolution II. Resolution to Amend the 19th ICLS Resolution Concerning Statistics of Work, Employment and Labour Underutiliza-*

*tion*. 21st. International Labour Organization.

International Labour Organization. 1982. *Resolution Concerning Statistics of the Economically Active Population, Employment, Unemployment and Underemployment, Adopted by the Thirteenth International Conference of Labour Statisticians (October 1982)*. No. 13th. Geneva, SWITZERLAND.

International Labour Organization. 2011. *Beyond the Measurement of Unemployment and Underemployment*. Https://www.ilo.org/publications/beyond-measurement-unemployment-and-underemployment.

International Labour Organization. 2026. *ILO Modelled Estimates Methodological Overview*.

International Labour Organization. n.d. *International Conference of Labour Statisticians (ICLS) - ILOSTAT*. Https://ilostat.ilo.org/about/standards/icls/?playlist=4194a13&video=38313ec.

Kale, Yemi, and Sani I. Doguwa. 2015. "On the Compilation of Labour Force Statistics for Nigeria." *CBN Journal of Applied Statistics* 6 (1(a)): 183–98.

Krishnamurthy, Shree Hari. n.d. *A Comprehensive Study of Applying Machine Learning Algorithms for Time Series Data Prediction to the Irish Labour Market Unemployment Rate*.

Morse, David A. 1970. "Unemployment in Developing Countries." *Political Science Quarterly* 85 (1): 1–16. https://doi.org/10.2307/2147555.

Mulero, Rodrigo, and Alfredo Garcia-Hiernaux. 2023. "Forecasting Unemployment with Google Trends: Age, Gender and Digital Divide." *Empirical Economics* 65 (2): 587–605. https://doi.org/10.1007/s00181-022-02347-w.

National Bureau of Statistics. 2016. *UNEMPLOYMENT/UNDER-EMPLOYMENT REPORT Q2 2016*. Q2 2016. National Bureau of Statistics (Nigeria).

National Bureau of Statistics. 2021. *Labour Force Statistics: Unemployment and Underemployment Report (Q4 2020)*. Q4 2020. National Bureau of Statistics (Nigeria).

National Bureau of Statistics. 2023. *Nigerian Labour Force Statistics Report Q4 2022 & Q1 2023*. Q4 2022 & Q1 2023. National Bureau of Statistics (Nigeria).

Ojapinwa, Taiwo, and Esan Folorunso. 2013. "Does Philips Relations Really Exist in Nigeria? Empirical Evidence." *International Journal of Economics and Finance* 5 (August): 123–23. https://doi.org/10.5539/ijef.v5n9p123.

Olukanmi, Seun O., Fulufhelo V. Nelwamondo, and Nnamdi I. Nwulu. 2021. "Utilizing Google Search Data With Deep Learning, Machine Learning and Time Series Modeling to Forecast Influenza-Like Illnesses in South Africa." *IEEE Access* 9: 126822–36. https://doi.org/10.1109/ACCESS.2021.3110972.

Pavlicek, Jaroslav, and Ladislav Kristoufek. 2015. "Nowcasting Unemployment Rates with Google Searches: Evidence from the Visegrad Group Countries." *PLOS ONE* 10 (5): e0127084. https://doi.org/10.1371/journal.pone.0127084.

Proshare Research. 2023. *Nigeria's New Unemployment Data in Numbers: Reality, Revision, and Reflection - Proshare*. Https://www.proshare.co/articles/nigerias-new-unemployment-data-in-numbers-reality-revision-and-reflection?menu=Economy&classification=Read&category=Nigeria

Richardson, S. C., Ralf Hussmanns, Farhad Mehran, and Vijay Verma. 1992. "Surveys of Economically Active Population, Employment, Unemployment and Underemployment." *The Statistician* 41 (2): 249. https://doi.org/10.2307/2348265.

Rob J Hyndman, and George Athanasopoulos. n.d. *Forecasting: Principles and Practice (2nd Ed)*.

Smith, Paul. 2016. "Google's MIDAS Touch: Predicting UK Unemployment with Internet Search Data." *Journal of Forecasting* 35 (3): 263–84. https://doi.org/10.1002/for.2391.

Tule, Moses, Taiwo Ajilore, and Godday Ebuh. 2016. "A Composite Index of Leading Indicators of Unemployment in Nigeria." *Journal of African Business* 17 (1): 87–105. https://doi.org/10.1080/15228916.2016.1113909.

Umoru, David, and M. A. Anyiwe. 2013. "Dynamics of Inflation and Unemployment in a Vector Error Correction Model." *Research on Humanities and Social Sciences* 3 (3): 20.

Vera-Jaramillo, Jaime. 2025. *Reconstructing Subnational Labor Indicators in Colombia: An Integrated Machine and Deep Learning Approach*. arXiv:2508.12514. arXiv. https://doi.org/10.48550/arXiv.2508.12514.

Wikipedia contributors. 2026. "List of Countries by Minimum Wage." *Wikipedia*, April.

World Bank. n.d. "Nigeria's Dichotomy: Low Unemployment, High Poverty Rates." In *World Bank Blogs*. Https://blogs.worldbank.org/en/opendata/nigerias-dichotomy-low-unemployment-high-poverty-rates.

Yi, Dingdong, Shaoyang Ning, Chia Jung Chang, and S. C. Kou. 2021. "Forecasting Unemployment Using Internet Search Data via PRISM." *Journal of the American Statistical Association* 116 (536): 1662–73. https://doi.org/10.1080/01621459.2021.1883436.

# 7 Appendix

Source code and extra materials can be found in this reprository.

Table 6: Wikipedia contributors (2026)

Minimum wage basis by country (Abridged)

| Country | Wage Basis | Notes |
|---|---|---|
| Afghanistan | monthly | Afs 5500/month for non-permanent private sector |
| Albania | monthly | L39087/month (~€471) |
| Algeria | both | DA 24000/month; DA 138.46/hour also published |
| Andorra | hourly | €7.70/hour |
| Angola | monthly | Kz 32181/month paid 13x per year |
| Antigua & Barbuda | hourly | EC$8.20/hour |
| Armenia | monthly | 75000 AMD/month (~$190) |
| Australia | hourly | Award-based; national minimum wage set per hour |
| Austria | none | No national minimum; collective bargaining by sector |
| Azerbaijan | monthly | 345 AZN/month (~$203) |
| Bahamas | hourly | B$5.25/hour (also daily/weekly equivalents) |
| Bahrain | none | Public-sector floor only; no private-sector statutory minimum |
| Bangladesh | monthly | 1500 BDT/month; industry-specific rates vary |
| Barbados | hourly | Bds$10.50-$11.43/hour by category |
| Belarus | monthly | Br 554/month (~$122) |
| Belgium | monthly | €2030/month |
| Belize | hourly | BZ$5.00/hour |
| Benin | monthly | CFA 52000/month |
| Bhutan | monthly | Nu 3750/month |
| Bolivia | monthly | Bs. 2500/month + Christmas bonus |
| Bosnia & Herzegovina | monthly | KM 543 net/month (~$269) |
| Botswana | hourly | P 7.34/hour (also monthly equivalent) |
| Brazil | monthly | R$1621/month paid 13x per year |
| Brunei | none | No statutory minimum wage |
| Bulgaria | both | 1077 BGN/month and 6.49 BGN/hour both official |

| | | |
|---|---|---|
| Burkina Faso | monthly | CFA 34664/month |
| Cambodia | monthly | KHR 818800/month (~$200) textiles sector |
| Cameroon | monthly | CFA 36270/month |
| Canada | hourly | Set per province/territory; federal floor also hourly |
| Cape Verde | monthly | 13000 CVE/month (~$141) |
| Central African Republic | both | FCFA 35000/month and FCFA 218.75/hour |
| Chad | both | FCFA 59995/month and FCFA 355/hour |
| Chile | monthly | CL$529000/month for workers 18-65 |
| China | both | Set locally; expressed monthly or hourly depending on region |
| Colombia | monthly | COP 1423500 + COP 200000 transport/month |
| Comoros | monthly | CF 55000/month |
| DR Congo | daily | FC 1680/day (~$1.83) |
| Republic of Congo | monthly | FCFA 90000/month (formal sector) |
| Costa Rica | daily | 11954-16000 CRC/day by sector |
| Cote d'Ivoire | monthly | CFA 60000/month (lowest category) |
| Croatia | monthly | €840/month |
| Cuba | monthly | $MN 2100/month (~$6.66) |
| Czech Republic | both | CZK 20800/month; CZK 124.40/hour |
| Denmark | none | No statutory minimum; collective bargaining (avg ~DKK 110/hr) |
| Djibouti | none | 2006 Labour Code replaced fixed rate with sector negotiations |
| Dominica | hourly | EC$4.00/hour |
| Dominican Republic | monthly | RD$8310-$15100/month by sector |
| Ecuador | monthly | $594/month (includes 13th & 14th salary components) |
| Egypt | monthly | Public sector: EGP 6000/month; private sector floor varies |
| El Salvador | monthly | $304.17/month |
| Equatorial Guinea | monthly | FCFA 129035/month |
| Estonia | both | €725/month; €4.30/hour |
| Eswatini | monthly | E 420-531/month by sector |
| Ethiopia | none | No national minimum; some state enterprises set own floors |
| Fiji | hourly | FJ$2.68/hour |

| | | |
|---|---|---|
| Finland | none | No statutory minimum; sector collective agreements required |
| France | both | €1801/month (official SMIC); €11.88/hour equivalent |
| Gabon | monthly | FCFA 150000/month |

Source: Data compiled from Wikipedia contributors (2026)

Table 7: International Labour Organization (n.d.)

# International Conference of Labour Statisticians (ICLS): Sessions and Key Resolutions

| Session | Year | Key Resolutions / What Was Introduced |
|---|---|---|
| 1st ICLS | 1923 | First international guidelines on labour statistics; basic definitions of employment and wages |
| 2nd ICLS | 1925 | Standards on wages and hours of work statistics |
| 3rd ICLS | 1926 | Guidelines on unemployment statistics; resolution on statistics of collective agreements |
| 4th ICLS | 1931 | Standards on statistics of industrial disputes and strikes |
| 5th ICLS | 1937 | Revised guidelines on wages and hours; led to ILC Convention No. 63 on wages and hours of work |
| 6th ICLS | 1947 | Post-war revival; guidelines on labour force statistics and employment measurement |
| 7th ICLS | 1949 | Standards on family living studies and household expenditure |
| 8th ICLS | 1954 | Revised definitions of employment and unemployment; guidelines on migration statistics |
| 9th ICLS | 1957 | Standards on statistics of occupational injuries and diseases |
| 10th ICLS | 1962 | Guidelines on labour cost statistics; revision of wages and hours standards |
| 11th ICLS | 1966 | Standards on measurement and analysis of underemployment and underutilization of manpower |
| 12th ICLS | 1973 | Revised integrated system of wages statistics; household income and expenditure survey guidelines |
| 13th ICLS | 1982 | Landmark resolution on the economically active population — modern definitions of employment, unemployment and underemployment |
| 14th ICLS | 1987 | Guidelines on occupational injuries statistics; statistics of collective bargaining |
| 15th ICLS | 1993 | First standards on statistics of employment in the informal sector |
| 16th ICLS | 1998 | Resolution on measurement of employment-related income; underemployment and inadequate employment; CPI guidelines |
| 17th ICLS | 2003 | Guidelines on informal employment (broadening 15th ICLS); CPI standards; called for ISCO-88 revision |
| 18th ICLS | 2008 | Resolution on child labour statistics; decent work measurement guidelines; amendment to 13th ICLS resolution |
| 19th ICLS | 2013 | Major overhaul — five forms of work; redefined employment as work for pay or profit; LU1-LU4 labour underutilization measures |

| | | |
|---|---|---|
| 20th ICLS | 2018 | Resolution on work relationships (ICSE-18); amendment to 18th ICLS child labour resolution; SDG indicator methodology |
| 21st ICLS | 2023 | Resolution on statistics of the informal economy; amendments to 19th, 16th, and 17th ICLS resolutions |

Source: Data compiled from International Labour Organization (n.d.) and ILO reports.